Is Irreligion correlated with increase in Leftism? Evidence from 23,000 Constituency-Years

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Abstract
Irreligion has been rising rapidly in Western democracies, and previous literature has linked Irreligion and Left-leaning political views at the individual level. However, given that many systems of government are based on electoral geographies, which are at the aggregate level, the link between Irreligion and Left-leaning political views is susceptible to the Modifiable Areal Unit Problem / Ecological Fallacy. To remedy this, I test whether increase in Irreligion is correlated with increase in Left-leaning vote shares in electoral constituencies in the United States, Australia, and England. My methodology is made up of three steps. First, for the United States, I estimate county-level religion estimates using Iterative Proportional Fitting. Second, for all countries, I spatio-temporally interpolate all variables onto election years in current electoral geography boundaries using population weighted areal interpolation and monotonic cubic splines. I then regress change in religion against change in Left-leaning party vote share using a spatial panel regression. I find that lagged Irreligion causes change in Left-leaning party vote share, but that differenced Irreligion has mixed results for Left-leaning party vote share. Thus, Irreligion does cause changes in electoral outcomes at the constituency level.

1. Background and Introduction

There have been various studies and books as of late on the remarkable increase in irreligion, coming from the fields of political science, religion, and anthropology. An important book in the space is *From Politics to the Pews* by Michele Margolis, which suggests rather counterintuitively that political attitudes preempt religious belief and cause changes in religious attendance and engagement with religious material. This idea of a life-cycle model of religion and politics in which a person develops social beliefs and religious ones simultaneously in youth is very different from how people have previously thought of religion and politics. Early religion and politics theorists put forth an 'ethnoreligious' model of religion and politics, in which individual religious communities behaved cohesively through mutual loyalty to the ends of the group (Smidt, Kellstedt and Guth, 2009).

This early model was challenged by another opinion, that churches in the United States were restructuring themselves around a conservative-liberal axis (Smidt, Kellstedt and Guth, 2009). This trend of conservative or liberal belief preempting and causing changes in religion is not new, as this restructuring model came into fruition in the eighties. However, this restructuring manifested itself not as leaving religion, but rather as going to a more liberal church. These beliefs have also been found to be important in more recent work on disaffiliation. For example, ethnographic research by Stephen Bullivant (2022) suggests that individual ire towards religious institutions perceived as traditionalist and conservative has much power in leading someone to leave church. He also finds that many people drift away from religion solely because they had originally been affiliated in order to appease family members (Bullivant, 2022). More descriptive work by Ryan Burge (2021) shines light on heterogeneity among Irreligious people, whom he dubs the 'nones,' after a common survey response that many

Irreligious people check. He finds that strong descriptors of Irreligious, such as 'Atheist' predict more liberal-leaning views, though Irreligious people on the whole are not deeply conservative or liberal in the aggregate. Taken together, this growing corpus shines light on the relationship between the largest-growing religious group in the West and political outcomes.

2. Objectives

There is reasonably strong evidence on the relationship between Irreligion and left-leaning politics. However, this relationship is theoretically complicated by the arrow of causality. Recent research suggests that political attitudes may crop up before religious ones, and that existing sentiments influence decision-making with regard to religious attendance propter hoc (Margolis 2018). Additionally, though much has already been made of the individual-level relationships between Irreligion and left-leaning politics, these works assume that a shift in society created by change in religion has some impact on an aggregate level. Of course, this falls to the ecological bias, as when aggregating up in geography, individual-level relationships may be non-existent or reversed based on heterogenous patterns within groups. Therefore, the effect that change in religion has on political society should be investigated at an aggregate as well as an individual level. Electoral geography is a natural example of a geography which could be deeply affected by moving religious trends. When I think of a staunchly conservative electorate, or a strongly progressive one, I often immediately consider the religious symbols of exclusion or tolerance that typify increasingly polarized campaigning.

Additionally, another factor that complicates much individual-level research is the central role that time plays in this narrative of Irreligion. Though various pieces of work include descriptive statistics with regard to time (Burge 2021), modeling relationships between a social trend and anything else requires more complex design that takes into account the temporal nature of social change. Thus, I produce a strong contribution to the existing work on the rise of Irreligion in political society by considering aggregate-level temporal trends and their effectiveness in influencing electoral geographies.

I hypothesize that an increase in Irreligion is associated with an increase in left-leaning vote shares in electoral districts.

Through this project, I hope to test my hypothesis on the relationship between Irreligion and Left-leaning political outcomes in electoral geographies. This means that the best outcome for this project is that I have a completed regression table with coefficients for Irreligion on Left-leaning political outcomes as well as maps and graphs showing off descriptives for my data.

3. Dataset and Methodology

I choose to analyze England and Wales, Australia and the United States because these countries have similar cultures, demographic makeups, voting systems, and similar declines in religion. Additionally, the censuses of England and Australia collect information on religion, which has mixed availability in other countries. The United States does not collect data on religion, so I create county-level estimates of religion using Iterative Proportional Fitting in the framework of Spatial Microsimulation. I then use a spatial panel model to regress change in religion against change in left-leaning party vote shares, controlling for various other shifts in population.

Type of Data	Source	Level/N	Year(s)
Demographics	U.S. Census	U.S. Counties, n = 3000	2000-2024
Microdata	National Annenberg Election Study	Individuals, n = 55,000	2000
Microdata	Pew Religious Landscape Study	Individuals, n = 35,000 per wave	2007, 2014
Microdata	UCLA Nationscape Study	Individuals, n = 474,000	2020
Demographics	Australian Census	Statistical Area 1, n = 61,000	1996-2021
Demographics	English and Welsh Census	Output Areas, n = 171,000	2001-2021
Election Results	MIT Election Data Science Lab	U.S. Counties, n = 3000	2000-2024
Election Results	eechdina Collected Australian House Results	House Districts, n = 150	2001-2025
Election Results	Pippa Norris' British Constituency Elections Results	House of Commons Constituencies, n = 575	2001-2024

The largest portion of the research design comes from imputing historical census and election data onto modern district geographies for election years. Because the shapes of parliamentary constituencies often change, I project census and vote data onto current parliamentary constituencies in two ways. For very small census geographies for which splitting geographies along constituency borders would not be a very large issue, I simply aggregate small geographies up to larger geographies using a representative point. This can be thought of as 'vertical' data creation. On the other hand, for larger geographies, such as previous election districts, I use a modified form of dasymetric interpolation which weighs population (from a satellite raster of population density) rather than amount of populated area. This can be thought of as 'horizontal' data creation in which modern districts are modeled as a population-weighted average of overlapping antecedent districts.

Imputation of census constituency-level estimates onto constituency-election-years are more rudimentary. I use the areal interpolation or aggregation scheme to project census data onto modern districts for census years then make intercensal estimates by interpolating between years with a monotonic cubic spline and extrapolating with linear regression. Intercensal estimates are often made with more complicated methods, such as aggregating births, deaths, and movement, but I don't have the sufficient microdata necessary to create more complex intercensal estimates. (I would like to do a more complex method here in a future version of this paper.)

Censuses in England and Australia contain data on religion, but these censuses are not frequent and, unlike the United States census, statistical agencies in these countries do not publish intercensal estimates (especially for religion). Additionally, vote counts in England are not reported at any level of geographic aggregation below the constituency level and vote counts in Australia are partially reported below the constituency level, but are completely tied to lower geographies because of the prevalence of mail-in and other voting methods, which are used in conjunction with compulsory voting. Given that in both, the drawing of districts is an apolitical process which preserves geography for most districts, using 'horizontal' interpolation to create vote shares is the best option.

To create county level estimates of religion, I use Iterative Proportional Fitting (IPF) to fit state- and regional-level survey data to demographic variables in counties. There are more complex methods of making small-area estimates of unobserved quantities, such as Multilevel-Regression and Post-stratification or ecological inference. However, I choose IPF because it does not require knowledge of joint distributions of observed variables and because its structure is not reliant on the sampling of the microdata. MRP-based methods, while more stable and probabilistic, require knowledge of individual cells of cross-tabulated observed variables, an approach that attempts to get around this constraint is largely contingent on the provenance of the sampling of microdata (Pitts et al. 2025). Other options, such as adjusting hierarchical regression with microdata and then independently imputing a portion of a cross-tabulation still requires knowledge of multi-variate joint distributions, for example, Kuriwaki, et al. (2024) made a joint distribution of five variables based on existing cross-tabulations with four and two dimensions. While these larger joint-distributions are available in more modern census data, they are not available in earlier estimates at the county level. (Especially given privacy implications with less populous geographic areas.)

However, IPF has serious drawbacks, of which the most obvious is the lack of uncertainty estimates. Because IPF is a deterministic matrix function, it is difficult to easily quantify error other than checking for deviations from observed variable marginals. Additionally, a cross-tabulation of multiple variables, in my case four, demands a lot from the microdata. For samples of 'only' twenty- to thirty-thousand, I had to aggregate up to broad regions in microdata to have sufficient responses for a four-variable cross-tabulation. This means that much variation within regions among demographic groups is lost. For example, I would expect other factors, such as urbanity or isolation or distinct minority groups in religious makeups of areas (e.g. Mormons, Pennsylvania Amish) to create much variation within cross-tabulated categories across regions. However, this granularity is lost. Even though this method is often called 'spatial' microsimulation, this framework does not actually incorporate any assumptions or weights to account for geographic autocorrelation.

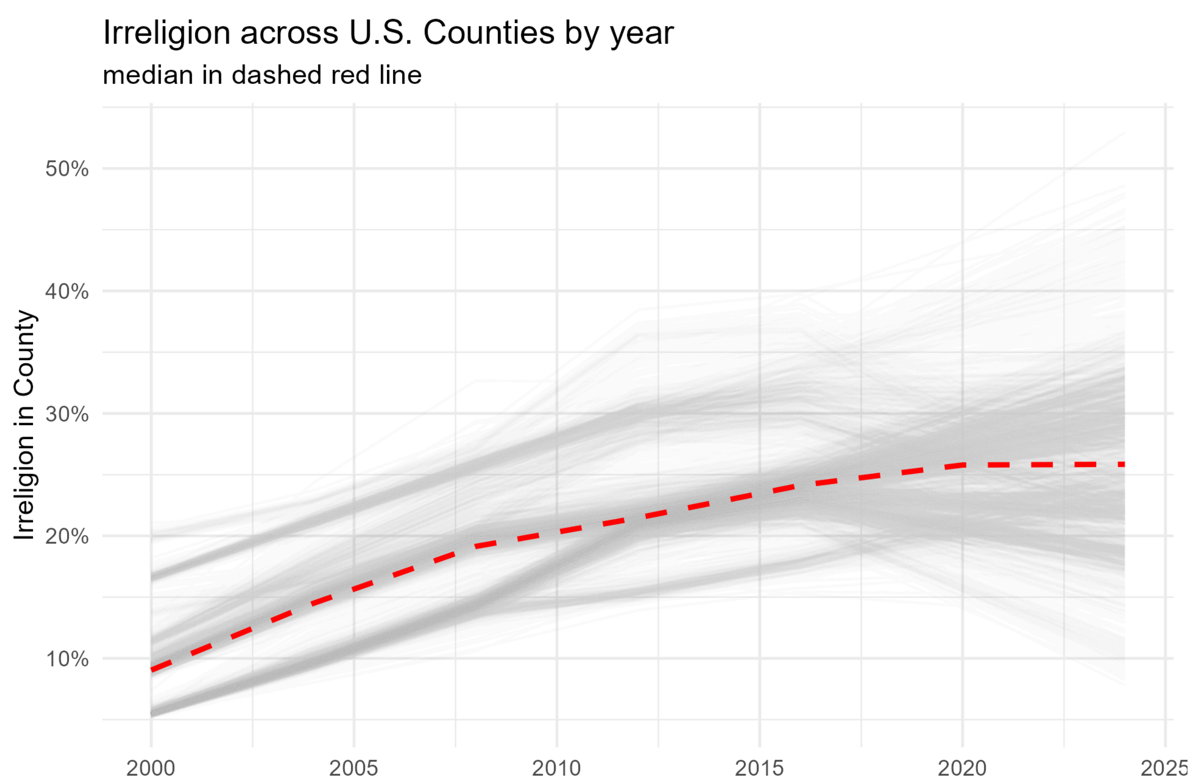
Despite these drawbacks, IPF is still the best solution given that other solutions would either be based entirely on the idiosyncratic sampling of individual surveys or require data that doesn't exist across all years. I choose to analyze counties, rather than congressional districts because many congressional districts are drawn in an uncompetitive manner and this politicization of district boundaries is heterogeneous across states. Counties, on the other hand, are apolitically drawn, unchanging (only one new county has been created in the last two decades), and also represent electoral constituencies.

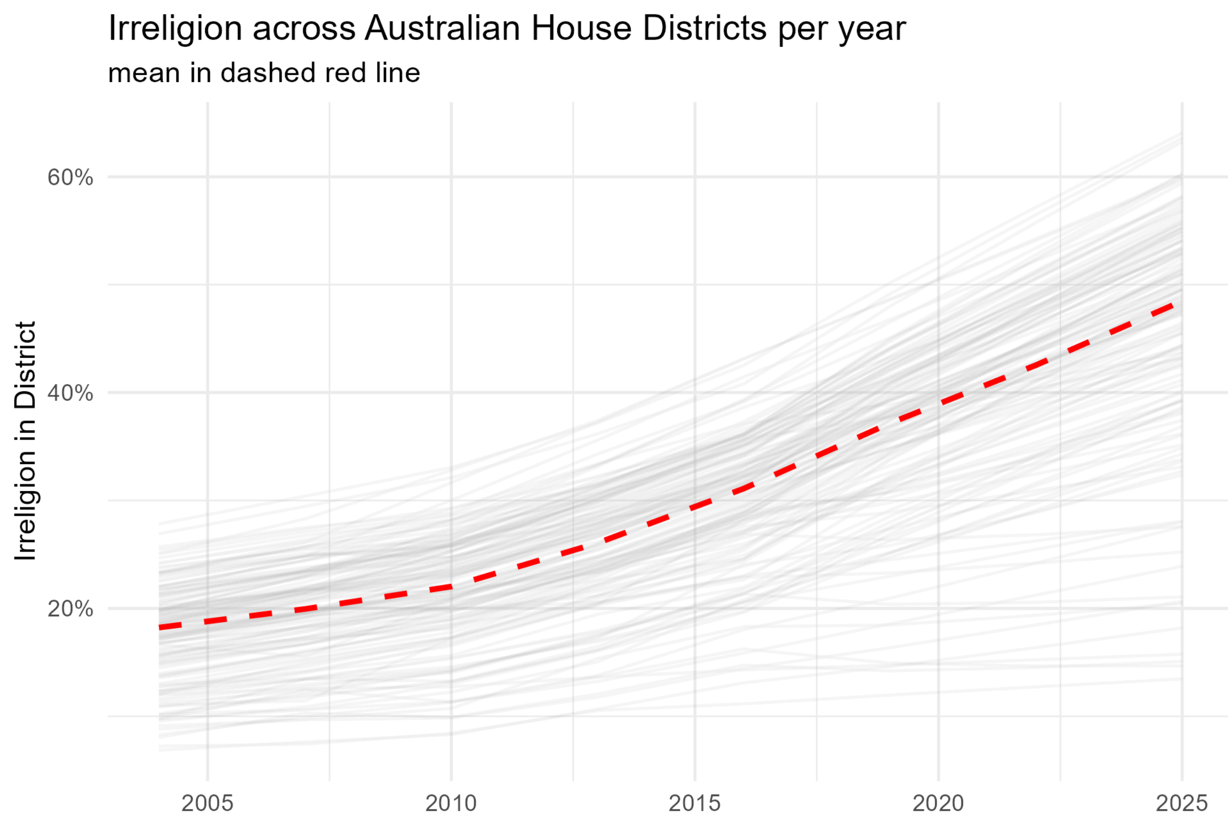
Each constituency is treated as a panel and the correlation between change in religion and change in left-leaning vote share is modeled as

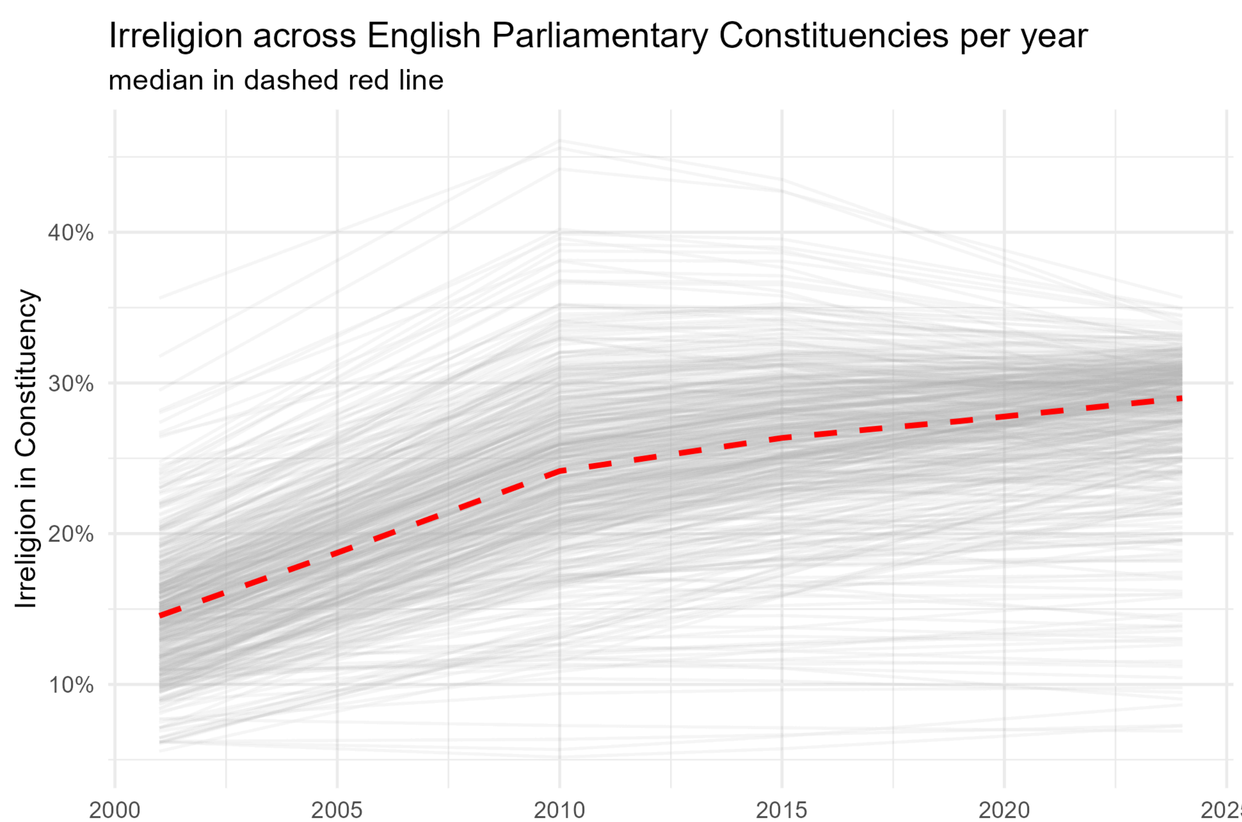
$$\Delta LeftVote = \beta_0 + \beta_1 \Delta Irreligion + \beta_2 Irreligion_{t-1} + \beta_{2k+1} \Delta Control_k + \beta_{2k+2} Control_{k,t-1} + \mu_i c_i + \rho W_y + \epsilon$$

in which change in the left-leaning vote in a geographic area is modeled as a function of lagged and differenced predictors, random effects, geographic autocorrelation, and error (Baltagi 2021). In all models, controls are comprised of some type of race/ethnicity, class/income, age, and education. Sex is not used as a control due to the general lack of variation in sex across geographic areas. I expect both coefficients on both differenced Irreligion and Lagged Irreligion to be positive, meaning that knowing where Irreligion was in a previous time period and knowing how much it has moved between then and the contemporaneous time period, are useful in predicting an increase in Left-leaning party votes shares in the contemporaneous time period. With permission, I use python for cleaning and modeling and R for graphing.

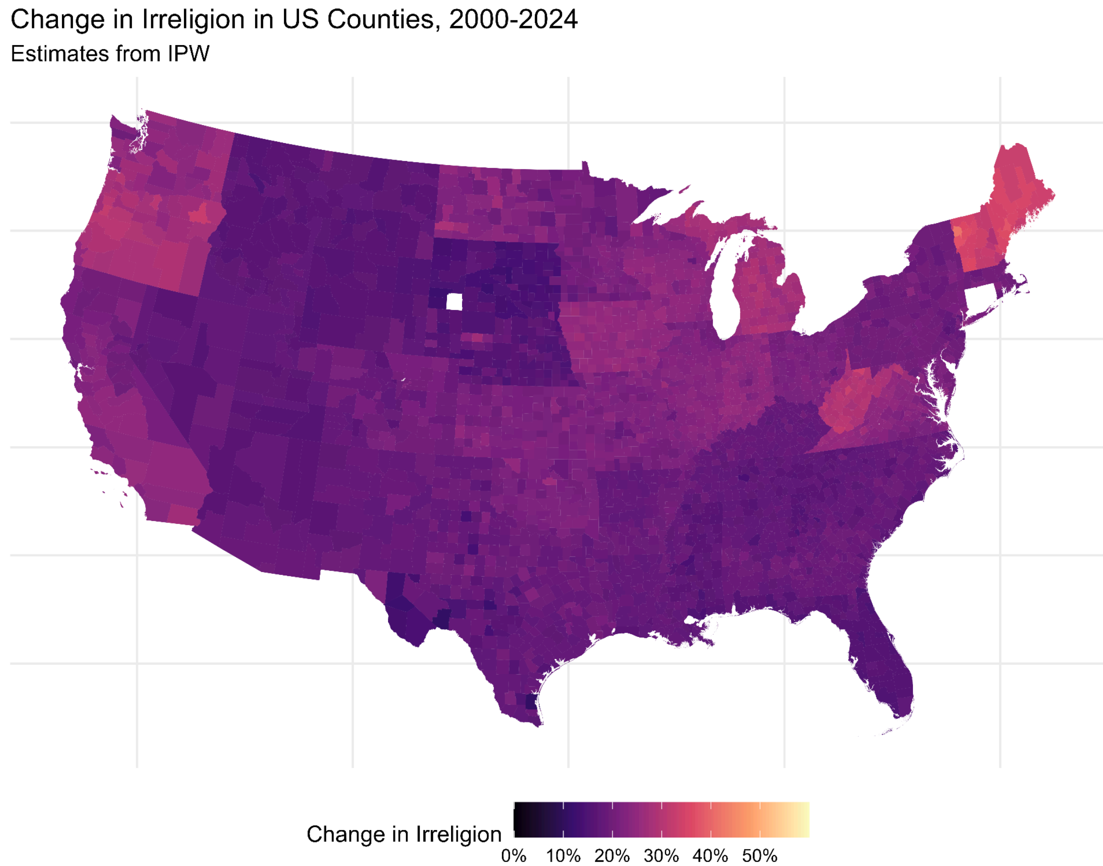
4. Results

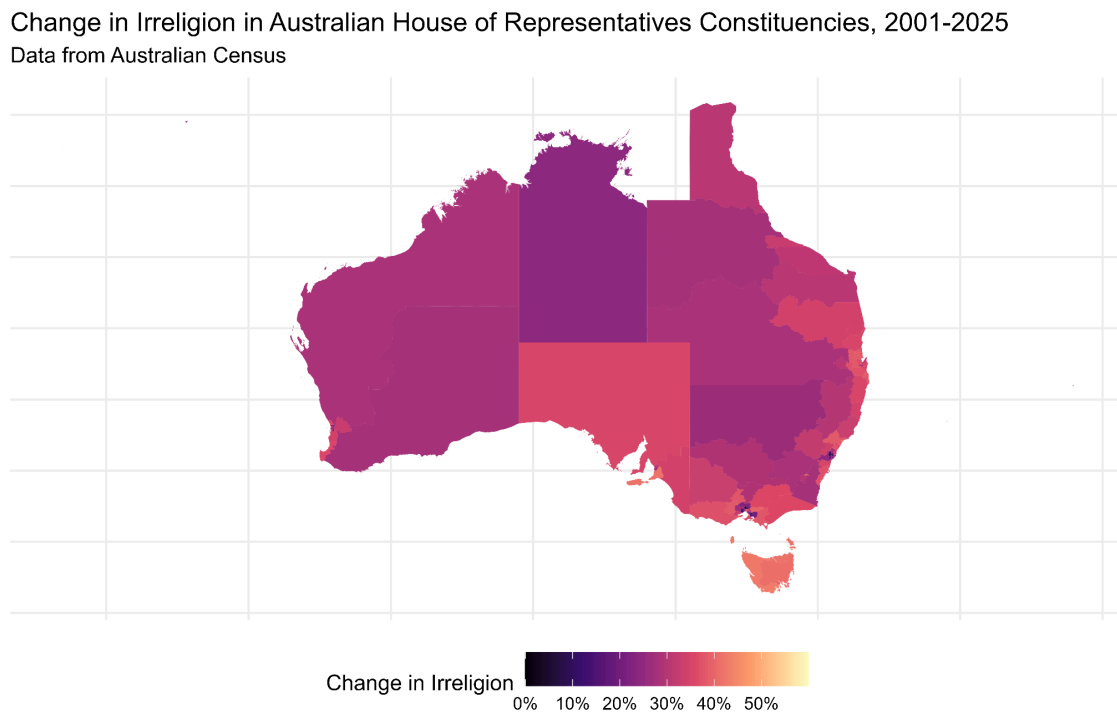


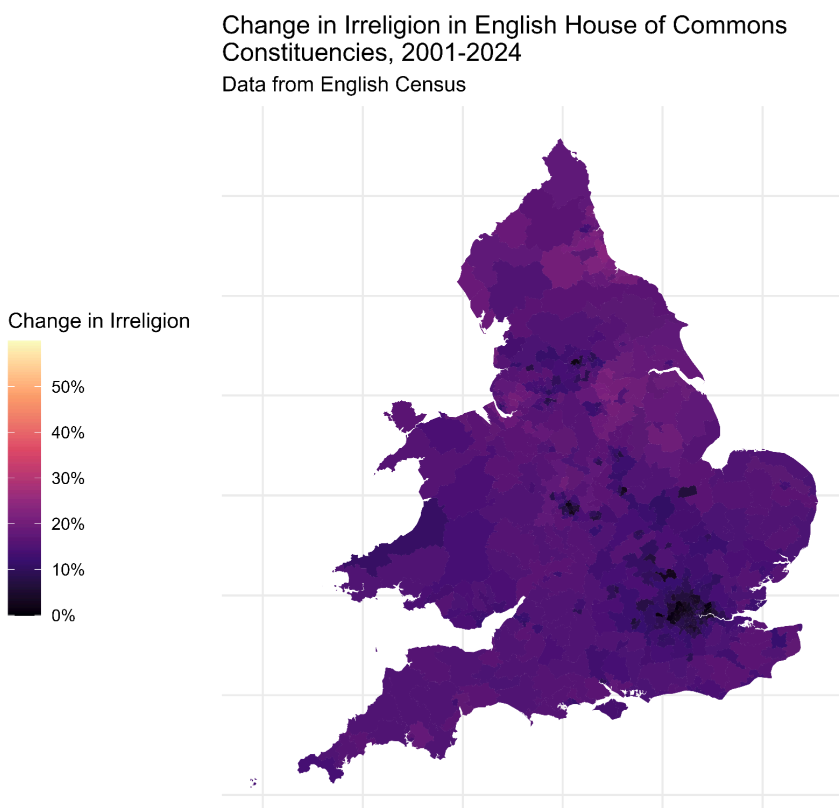

Irreligion across Australian House Districts per year
mean in dashed red line
Irreligion in District
60%
40%
20%
2005
2010
2015
2020
2025


Irreligion across English Parliamentary Constituencies per year
median in dashed red line
Irreligion in Constituency
40%
30%
20%
10%
2000
2005
2010
2015
2020
2025

Starting with descriptives, Irreligion is generally rising across the world, though it appears to be plateauing in electoral geographies in the United States and England. The median United States county is 26% Irreligious, up from 8% in the early 2000's. Irreligion in Australian and English parliamentary constituencies are comparatively higher, with Australia reaching nearly 50% Irreligious the median house district, with some in the 60s. This is again, a remarkable increase that demonstrates seismic shifts in the demographics of electoral geographies. There is, however, vast variation, especially geographic, in this trend. The United States counties map demonstrates some difference between regions, with cities in the Northeast and the West Coast being most likely to have high irreligious populations, whereas in rural and highly religious areas such as the Rockies, religion appears to have slipped very little. As the map of Australia shows, Australia is more advanced along the trend of religious disaffiliation, with even rural areas becoming Irreligious.




Change in Irreligion in Australian House of Representatives Constituencies, 2001-2025
Data from Australian Census
Change in Irreligion
0%
10%
20%
30%
40%
50%


Change in Irreligion in English House of Commons Constituencies, 2001-2024
Data from English Census
Change in Irreligion
50%
40%
30%
20%
10%
0%

	Spatial Lag Models		
	United States	England	Australia
$\Delta_{irreligion}$	0.176*** (0.01)	0.151 (0.09)	-0.503*** (0.11)
$Irreligion_{t-1}$	0.079*** (0.01)	-0.240*** (0.04)	0.178*** (0.03)
ρ	0.135*** (0.00)	0.343*** (0.02)	0.102*** (0.02)

As can be seen from the lagged models, my hypothesis finds mixed support. The United States counties demonstrate strong electoral behavior that reflects religious change in society. As expected, both differenced and lagged irreligion are significantly useful in predicting left-leaning vote shares in electoral districts. However, the English and Australian models confound this somewhat. As those models are likely more valid, the positive result that the US model shows may be spurious or rather complicated. In the English model, the coefficient of differenced irreligion is positive whereas that of Australia is negative. The opposite effect is seen for the coefficient on lagged irreligion. I initially thought that these two coefficients were competing with each other and thus overfitting to the data, hiding a null result. Upon removing the differenced variables, the negative coefficient on lagged irreligion in England remained, as did the positive coefficient for lagged irreligion in Australia. The trends for demographic variables are far slower moving and less chaotic than change in two-party preferred vote shares in Australia or voting for left-leaning parties in England. However, this heterogeneity between units may also suggest that the effect of irreligion on the politics of different countries may itself be heterogeneous. Irreligion is evidently useful in all models for predicting some version of left-leaning vote share, so I doubt the predictor itself is entirely spurious.

Of course, error from different data sources and from multiple levels of imputation may also be affecting the outcome of the spatial lag models. Clear discontinuities can be seen in my county-level religion estimates from the boundaries of the regions which the microdata were split into. I regret that I wasn't more able to incorporate increased errors into the model.

5. Conclusions

This paper, despite its mixed results, demonstrates that at the very least, something is happening at an aggregate level of partisan preference when masses of individuals move away from religion. As Irreligion increases, Left-leaning vote share also tends to increase, at the level of the electoral constituency. There is, however, a lot of noise. A limitation of this study is that the temporally imputed data is constant in slope between census years, which means that differenced terms are not very different from each other. This means that the current models catch a small amount of signal and a lot of noise. Lower resolution temporal data could remedy this. Additionally, in a future version of this paper, I would like to add data from Canada and New Zealand. I think that this paper contributes an expected but important finding to the growing literature on Irreligion in Western societies. Now that it has been somewhat established that Irreligion is involved in constituency-level elections, maybe projects or synthetic control counterfactual elections would be a good avenue for future research.

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